

A Discrete Framework for the Structural Taxonomy of 2D Jordan Curves via Radial Manifold Deformation and Midpoint Caustic Analytics

Framework Environment mx23
Independent Research Group, Desktop01 Linux Cluster

June 27, 2026

Abstract

We present a novel, discrete computational framework for analyzing the configurations of inscribed polygons within arbitrary 2D Jordan curves, focusing on Toeplitz’s conjecture (the Inscribed Square Problem). Traditional analytical methods depend heavily on boundary differentiability (C^2 continuity), which breaks down when confronted with rigid polygonal junctions, non-convex reflex corners, or non-differentiable fractals. To bridge this gap, we introduce the Central Chord Radial Transform (CCRT), a coordinate-free metric descriptor that processes discrete boundary distributions into unoriented internal chord arrays. By investigating the structural variance of these arrays under spatial translation, we establish the existence of a strictly convex parabolic potential well, enabling autonomous center-of-mass recovery via numerical gradient descent. Furthermore, we analyze the configuration space of the midpoint fields, demonstrating that they map 2D projections of an immersed, non-orientable Möbius manifold. Finally, through Chaikin corner-cutting and high-frequency sinusoidal stress tests, we show that periodic symmetric loops function as geometric resonators, generating nested, self-similar families (clones) of inscribed objects.

1 Introduction

The Inscribed Square Problem, posited by Otto Toeplitz in 1911, asserts that every continuous simple closed curve (Jordan curve) in the plane contains four distinct vertices forming a perfect geometric square. While proven for sufficiently smooth curves (e.g., piecewise analytic or convex profiles), the conjecture remains unproven for general continuous loops. Most topological literature relies on non-constructive, top-down proofs using algebraic topology and intersection theory. In this paper, we develop a bottom-up, data-driven computational pipeline designed to construct, visualize, count, and classify these embedded

configurations across highly irregular topological domains.

2 Methodology: The CCRT Engine

Given a discrete Jordan curve γ discretized into N boundary coordinate vectors $x_i \in \mathbb{R}^2$, we establish an internal tracking origin $C = (x_C, y_C)$. The CCRT executes a ray-casting transformation: for every sampled vertex x_i , a ray is cast through C to find its unique intersection with the opposing boundary profile $y_i \in \gamma$:

$$y_i = \arg \min_{z \in \gamma} \|z - (2C - x_i)\| \quad (1)$$

This transform extracts a discrete universe of structural chords characterized by an ordered dataset of point pairs $\{x_i, y_i\}$. This dataset yields two critical secondary metrics:

1. **Chord Length (L_i):** $L_i = \|x_i - y_i\|$
2. **Chord Midpoint (M_i):** $M_i = \frac{x_i + y_i}{2}$

3 Emergent Properties of the Midpoint Manifold

3.1 The Variance Potential Well and Autonomous Centering

A fundamental breakthrough of this framework is the behavior of the total structural variance (σ_M^2) of the midpoint dataset relative to a translated tracking origin C :

$$\sigma_M^2 = \text{Var}(M_x) + \text{Var}(M_y) \quad (2)$$

Empirical tracking confirms that σ_M^2 forms a strictly convex, parabolic potential well across the 2D plane. If C perfectly coincides with a curve’s native axis of central symmetry (e.g., a circle), information entropy collapses completely into a single Dirac delta point singularity, where $\sigma_M^2 = 0.000000$. We exploit this property by constructing

an autonomous self-centering robot that optimizes coordinate placement using a numerical gradient descent update loop:

$$C_{t+1} = C_t - \eta \nabla \sigma_M^2(C_t) \quad (3)$$

Where η represents the learning parameter, driving the origin dynamically into the basin of minimal entropy.

3.2 Topological Projections of the Möbius Strip

In low-dimensional topology, the configuration space of all unordered pairs of distinct points on a closed loop forms an open Möbius strip. By tracking the unordered midpoints $\{M_i\}$ under a globally optimized frame of reference, the resulting **Midpoint Caustic Cloud** acts as a literal 2D immersion of this non-orientable surface. The high-density lines within the cloud (caustics) depict catastrophe folds where the geometric properties of the curve concentrate.

4 Numerical Scaling and Discretization Dynamics

To evaluate the stability of the configuration search routine, identical bumpy Bezier loops were swept across a spectrum of discretization resolutions ($N = 100$ to $N = 1200$). The results expose a profound grid-aliasing phenomenon:

- **Coarse Grids** ($N = 100$): High informational noise. Wide tolerance filters capture slightly warped trapezoids, returning a high count of false-positive “ghost” squares.
- **Optimal Balance** ($N = 300$ to 600): The data plateaus and stabilizes at exactly 2 independent inscribed squares, identifying the precise geometric signature of the loop.
- **Ultra-Fine Grids** ($N = 1200$): Numerical starvation occurs. If search filters are tightened lineally alongside grid density, the infinitesimal gaps between adjacent discrete coordinates cause the perpendicular chord intersections to step directly past one another, dropping the recorded square count to 0.

5 Extreme Stress Testing Matrix

5.1 Rigid Polygons vs. Chaikin Corner-Cutting

The Triangle Boundary: Stiff equilateral vertices generate a crisp 3-spoked Y-triskelion caustic web.

The Hexagonal Bifurcation: A Regular Hexagon forces a highly constrained midpoint distribution, collapsing entirely into a perfect central dot, where its internal symmetries limit orthogonal chord alignment. However, deforming it into an Irregular Hexagon fractures the point

singularity into an asymmetric cloud, instantly unlocking hidden topological degrees of freedom that isolate multiple authentic squares.

Chaikin Refinement: By replacing sharp vertices with smooth, corner-less transitions using Chaikin’s Corner-Cutting Algorithm (a 20% edge slicing depth), we simulate quadratic Bezier junctions. This de-stiffens the linear caustic spokes into fluid curves, eliminating lattice-aliasing while preserving the global symmetry group.

5.2 Deep Non-Convexity: The L-Shape Ghost Manifold

Subjecting an orthogonal L-shaped bracket containing a 270° reflex corner to the CCRT engine uncovers an extraordinary geometric event: **The configuration manifold physically detaches from the shape interior.** Because the shape folds sharply inward, the straight lines connecting opposite walls must cross outside the boundary. The framework projects a smooth, curved “ghost caustic ribbon” directly into empty space outside the shape walls.

5.3 Periodic Sinusoidal Starbursts: The Clone Generator

When evaluated against a high-frequency, 12-peak sinusoidal starburst curve ($r = 0.8 + 0.3 \sin(12\theta)$), the midpoint cloud explodes into a complex geometric web. Our findings confirm that periodic symmetric curves act as geometric resonators. They establish discrete, radial concentric thresholds where squares replicate internally—generating nested, self-similar families (clones) of inscribed objects targeting outer peaks and inner valleys simultaneously.

6 Conclusion

This project demonstrates that global topological features of complex, non-smooth, and non-convex curves—specifically the guaranteed tracking and inscription of Toeplitz configurations—can be mapped, counted, and classified cleanly via discrete algorithmic computation. By treating chords as informational straws and utilizing midpoint caustics as shape descriptors, we establish an elegant, empirical taxonomy for 2D Jordan curves.

References

- [1] O. Toeplitz, “Ueber einige Aufgaben della Geometria situs,” *Verhandlungen der Schweizerischen Naturforschenden Gesellschaft*, 1911.
- [2] G. Chaikin, “An algorithm for high-speed curve generation,” *Computer Graphics and Image Processing*, 1974.
- [3] H. Vaughan, “The Inscribed Rectangle Problem,” *American Mathematical Monthly*, 1981.

- [4] T. Tao, “An integration approach to the Toeplitz square peg problem,” *Experimental Mathematics*, 2016.